

SRUL: Spatial Spherical Latents with Riemannian Flow Matching

Match the prior to the latent support learned by the autoencoder

Motivation

An autoencoder does not have to produce Gaussian latents

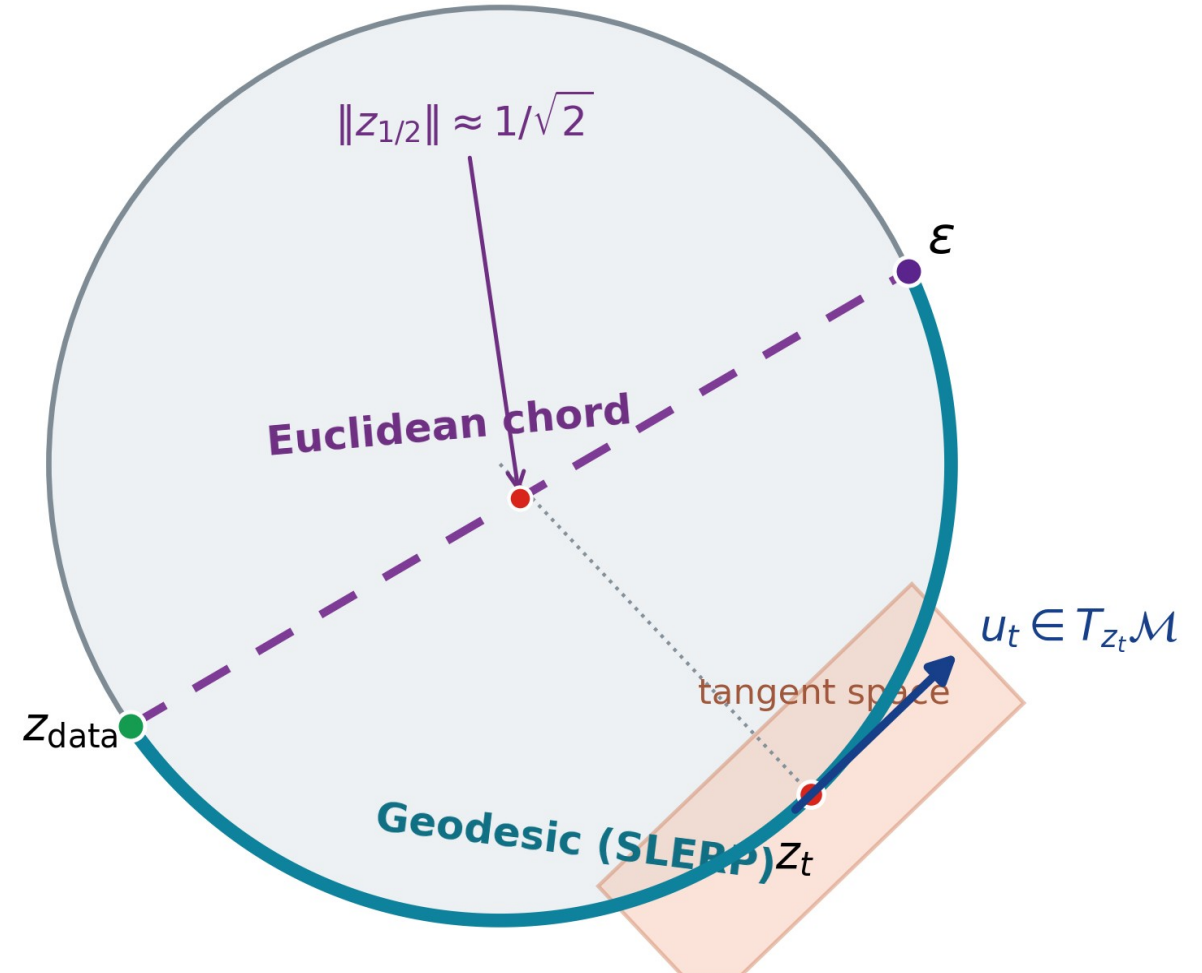
Normalization can make token norms nearly constant (e.g., Transformer + LayerNorm)

The prior can mismatch the latent support or leave it along the transport path

Main contribution

Train spherical tokens and RFM together then test path and support mismatch

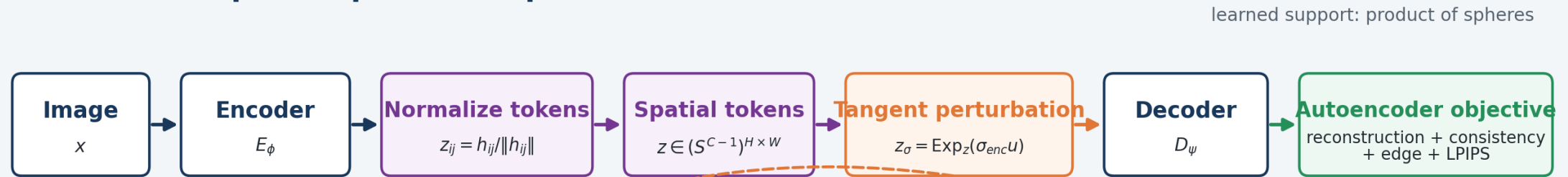
CIFAR-10 • PathMNIST • CelebA-64



SRUL pipeline

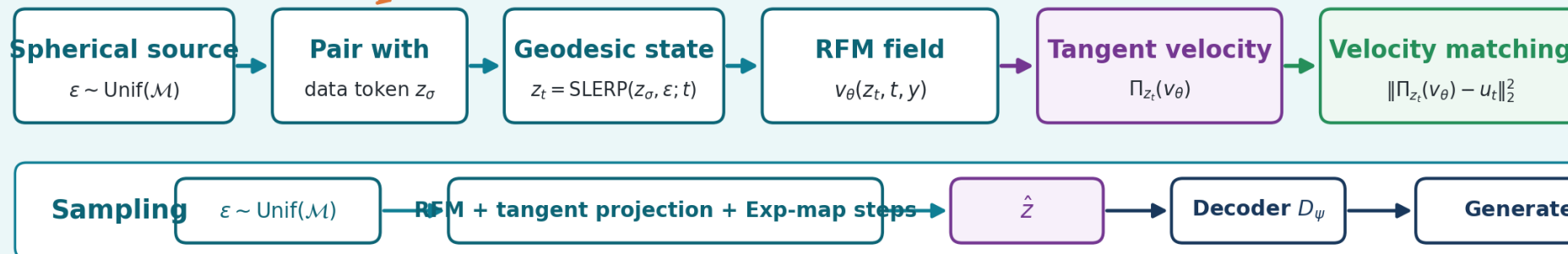
Learn the spherical representation, learn its distribution, then sample on the same manifold

A. Learn the spatial spherical representation



B. Learn and sample the latent distribution with RFM

dashed arrow: encoder targets used to train the prior

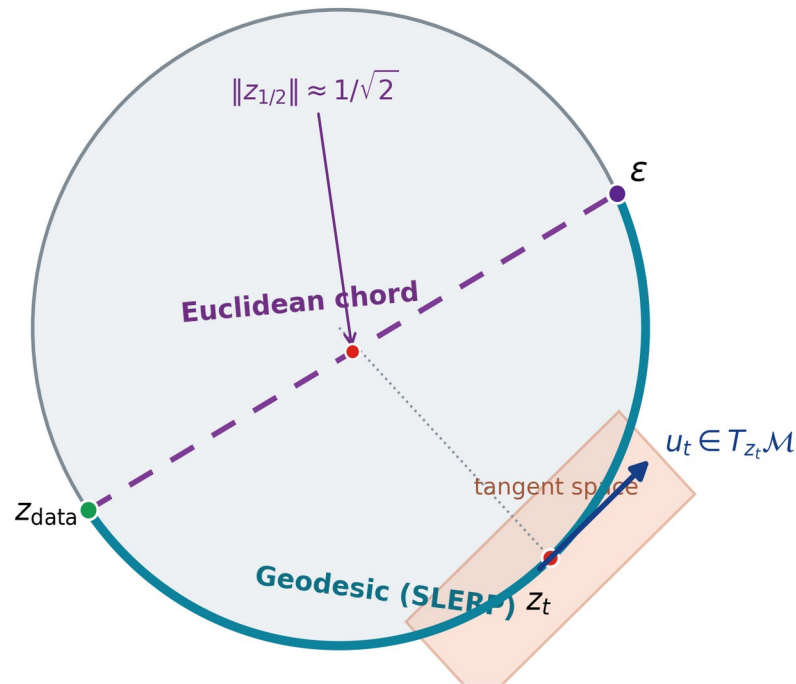


Core geometry: token normalization + tangent perturbation + SLERP + tangent projection + exponential-map integration.

Why the autoencoder and prior must match

Two controlled tests separate path mismatch from support mismatch

Same spherical endpoints, different transport path



1. Path match: same spherical autoencoder

Method	FID	KID	Prec.	Recall	Min norm
Chord	67.44	0.0740	0.840	0.085	0.705
SRUL	63.84	0.0667	0.857	0.091	1.000

2. Support match: SRUL vs. the same VAE with two priors (CFG = 2)

Method	FID	Prec.	Recall
SRUL: spherical AE + RFM	49.42	0.857	0.110
VAE + standard FM	50.70	0.855	0.091
VAE + projected RFM	56.04	0.848	0.063

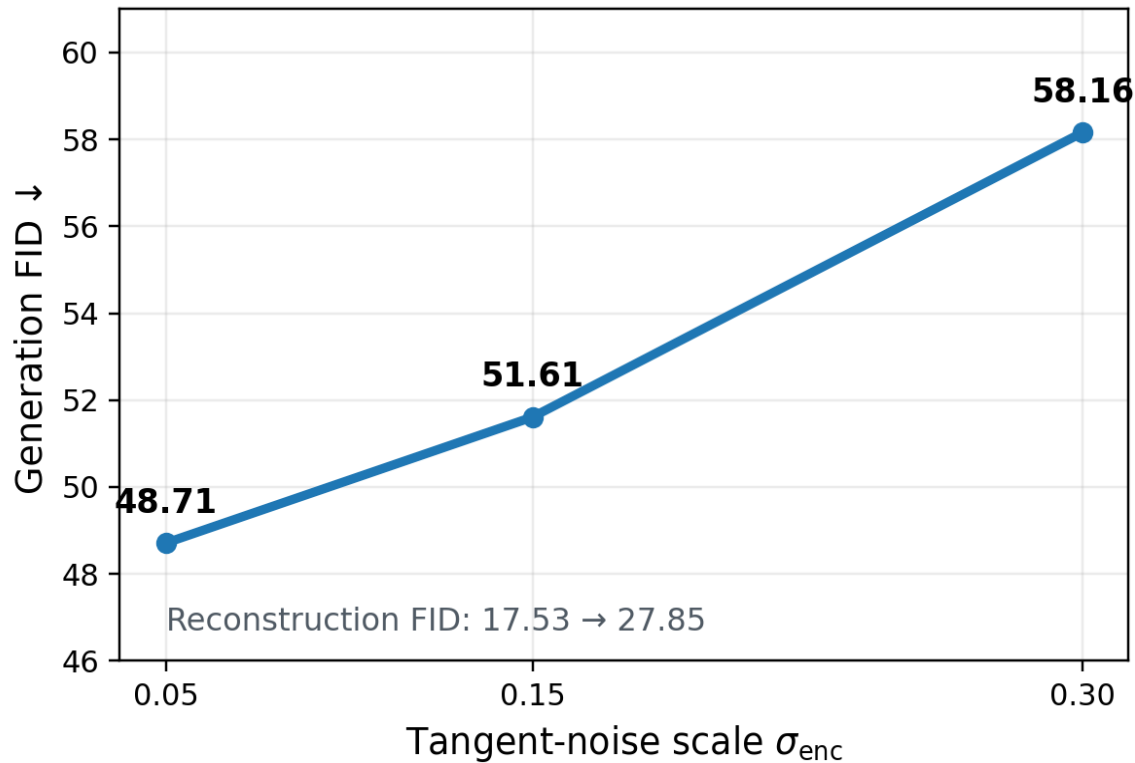
Same VAE: radius CV = 0.291; mean-radius replacement changes reconstruction FID 16.90 $\rightarrow$ 22.46.

SRUL learns spherical support before RFM. Projecting a VAE after training removes radius information.

Two controls after the geometry is chosen

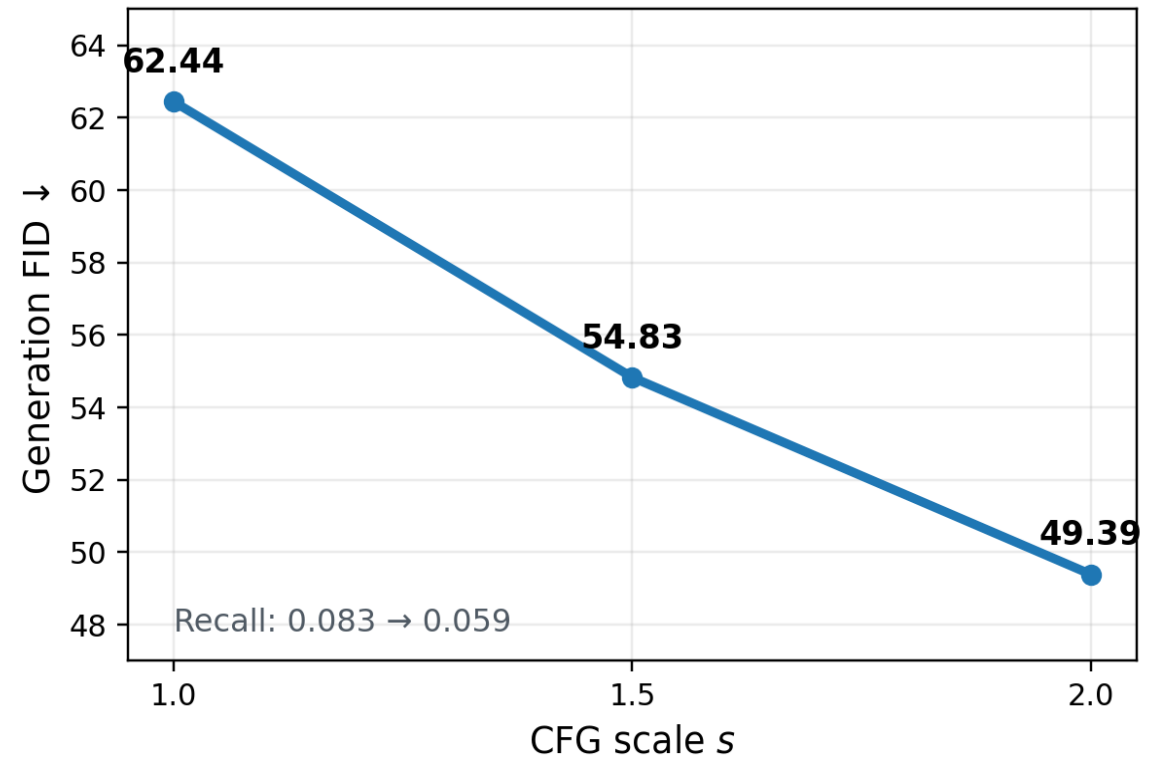
Tangent noise changes the latent representation; classifier-free guidance changes sampling

Noise ablation



Tangent-noise sweep: $\sigma_{\text{enc}} = 0.05$ gives the best tested reconstruction and generation quality.

Guidance ablation



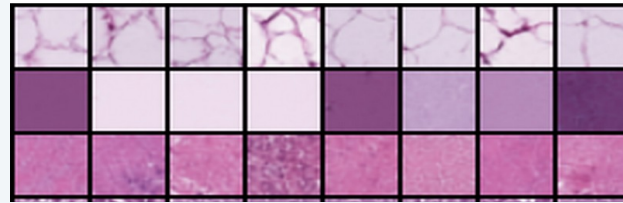
CFG sweep: stronger guidance improves FID and precision, while recall decreases.

SRUL in three image settings

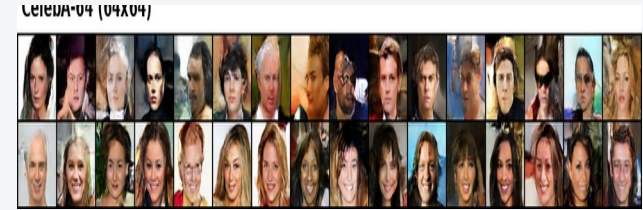
The same method is tested on different image structure and resolution



CIFAR-10



PathMNIST



CelebA-64

Dataset	Evaluation setting	Grid	Recon. FID	Gen. FID	Prec.	Recall
CIFAR-10	Geometry + sampling ablations	4×4	17.53	48.71	0.861	0.090
PathMNIST	Histopathology transfer	4×4	7.13	29.36	0.829	0.309
CelebA-64	64×64 face generation	8×8	9.08	41.22	0.657	0.198

CIFAR-10 tests geometry and controls; PathMNIST tests texture transfer; CelebA-64 tests 64×64 faces.